

Lucy Lai

Harvard Program in Neuroscience PhD Statement of Purpose

As a neuroscientist and future professor, I plan to combine experiments and theory to elucidate the neural principles underlying sensorimotor integration in health and disease. By pursuing a Ph.D. in Neuroscience, I seek to acquire the tools necessary to conduct rigorous experimental investigations of neural circuits and to discover the computational principles that govern circuit function in complex behaviors.

Inspired by my thirteen years of training as a classical pianist, I became fascinated with how the brain processes complex perceptual scenes and learns sophisticated motor sequences. Curious as to how our brains guided our interactions with the world, I declared a major in Cognitive Sciences and joined Dr. A's lab at School A in my freshman year to investigate how brains processed sensory information. In the A Lab, I designed and implemented a psychophysical experiment where human subjects made duration judgements on multisensory stimuli to investigate how the brain uses context to inform multisensory perception. I found that the same multisensory cues produced different percepts depending on the context in which they were presented. With data in hand, I recognized the need to gain experience in computational modeling of behavior, as no previous model in the multisensory literature could explain the pattern of interactions that we had observed.

Seeking an opportunity to learn more about computational neuroscience, I applied to and was awarded a travel grant to attend the 2016 Computational and Systems Neuroscience (Cosyne) conference in my sophomore year. Inspired by the collaborative work of theorists and experimentalists and convinced that theory unified principles of neural computation, I began adding quantitative courses in mathematics, statistics, and computer science to my curriculum in hopes of applying theoretical approaches to my research.

At Cosyne I met Dr. B, whose work blended behavior and computational modeling to characterize the brain processes underlying sensorimotor timing. The summer after my sophomore year, I joined Dr. B's lab at School B as part of an NSF-REU program to investigate how timing is represented in memory. I designed and implemented a psychophysical experiment where human subjects recalled and produced time intervals from memory and developed a Bayesian inference model that could explain subjects' bias and variance in interval timing. I presented my work as a poster and talk at Conference B, and have continued working with Dr. B to prepare a first-authored manuscript of my work. With the skills I gained from School B, I was better equipped to engage my research back in the A Lab.

During my junior year, I resumed my exploration of multisensory perception in the A Lab. After nearly a year of trial-and-error, I developed a Bayesian inference model that characterized how multisensory cues interact in different contexts and hypothesized that perceptual outcomes are conditioned on decisions to bind or segregate sensory cues. With the support of a X grant, I presented posters of my work at Conference A1 in 2016, Conference A2 in 2017, and Conference A3 in 2017. Additionally, I wrote my Neuroscience honors thesis on this research, and am about to submit a first-authored manuscript of my work to a peer-reviewed journal.

Behavioral psychophysics and computational modeling provide methods to study perception and sensorimotor action in humans. However, I believe that a comprehensive understanding of brain function necessitates investigation of the underlying neural mechanisms that drive complex behaviors. Seeking to learn neurophysiology, I joined Dr. C's lab at School C as part of the

Undergraduate Scholars program to probe the neural representations underlying action selection. By recording extracellularly from neurons in the motor cortex and striatum of mice performing a directional reaching task, I found that while both areas represented action choice, striatum selects the movement parameters, such as velocity, of the action. At the end of the summer, I presented my work as a poster and talk to the C community, and am now preparing a figure of data from my neural recordings to be incorporated into a manuscript.

My experiences at A, B, and C equipped me with the skills necessary to articulate my research questions and to design robust experiments to answer them. More fundamentally, I have learned that research is both challenging and fulfilling. Despite the reality of failed experiments and the levels of persistence and patience required in research, I remain driven by the satisfaction that comes from inquiry and discovery. I seek to continue pursuing my curiosities in Harvard's Program in Neuroscience, where I plan to build a broad, interdisciplinary skillset for conducting neuroscience research at different levels of explanation.

Vibrant labs across the multiple PiN-affiliated departments and hospitals provide exceptional environments for the kind of graduate training I seek. In particular, Dr. Alpha's research resonates with my interest in using a range of behavioral, imaging, and computational approaches to understand the cortical microcircuits involved in sensorimotor decision making. While I have had some experience in neural data analysis, I am eager to use and develop new approaches for characterizing and understanding the dynamics in cortical circuits that collectively support complex behaviors. I am also interested in Dr. Beta's research in motor sequence learning, and in particular how neural algorithms code for the generation and regulation of motor variability. Finally, I would be interested in collaborating with theorists such as Dr. Gamma or Dr. Epsilon to sustain a dialogue between experimental data and the theoretical predictions of neural models. If accepted to PiN, I plan to rotate through these labs; to this end, I have already had extensive interactions and communication with Drs. Alpha and Gamma regarding our mutual research interests and their ongoing work.

With access to the Center for Brain Science, the Longwood Medical Center, and other Harvard-affiliated hospitals, PiN is uniquely positioned to provide interdisciplinary training and opportunities for collaboration. I intend to make the most of this enriching environment by collaborating with colleagues across disciplines. As my next step in training, Harvard's Program in Neuroscience will equip me with the skills and resources to conduct rigorous investigations of nervous system function that will reveal how the brain gives rise to behavior, and how knowledge of the underlying mechanisms can be leveraged to improve human experiences compromised by disease.